THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Assertion (A): Chromatography can separate the colours of a black ink. Reason (R): Different dyes have different solubilities in the solvent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): The mass of a gas-releasing reaction in an open vessel appears to decrease. Reason (R): Mass is destroyed during reactions in open vessels. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Isotopes of an element have similar chemical properties. Reason (R): Isotopes have the same number of electrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): A body moving in a circle with constant speed is accelerated. Reason (R): The direction of its velocity changes continuously. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A 5 kg block on a smooth horizontal floor is pulled by a horizontal force of 20 N. Its acceleration is: 1

(a) 5 m/s squared **(b)** 4 m/s squared **(c)** 2 m/s squared **(d)** 20 m/s squared

6. A person of mass 60 kg runs up a flight of 20 stairs, each 20 cm high, in 5 s. Power is (g = 10): 1

(a) 480 W **(b)** 240 W **(c)** 120 W **(d)** 960 W

7. A sonar signal gives echoes from two surfaces at 0.8 s and 1.0 s. The separation between the surfaces (v = 1500 m/s) is: 1

(a) 300 m **(b)** 75 m **(c)** 600 m **(d)** 150 m

8. The structure that gives the plant cell its definite shape is the: 1

(a) cell membrane **(b)** vacuole **(c)** cell wall **(d)** nucleus

9. The reflex action of removing the hand from a hot object involves: 1

(a) muscle only
(b) bone only
(c) blood only
(d) sensory neuron, relay neuron and motor neuron

10. A scientist finds an organism that has no cell wall, is multicellular, has a body with tissues, and feeds on other organisms. It is a(n): 1

(a) animal **(b)** protist **(c)** plant **(d)** fungus

11. Seeds dispersed by wind have: 1

(a) hooks **(b)** fleshy covering **(c)** wings or hairs **(d)** no coat

12. The troposphere is warmer near the ground mainly because: 1

(a) the ground is heated by sunlight and warms the air
(b) the Sun is closer
(c) greenhouse gases are absent
(d) ozone absorbs UV

13. A metre scale has a worn end and reads 0.3 cm at the start. A measured length reads 5.7 cm. The corrected length is: 1

(a) 0.3 cm **(b)** 5.7 cm **(c)** 6.0 cm **(d)** 5.4 cm

14. The solubility of KNO3 is 64 g/100 g water at 40 degrees Celsius and 32 g at 20 degrees Celsius. 164 g of saturated solution at 40 degrees Celsius is cooled to 20 degrees Celsius. The mass of crystals is: 1

(a) 32 g **(b)** 96 g **(c)** 64 g **(d)** 16 g

15. A sample of a compound contains 2.4 g of carbon and 6.4 g of oxygen. The ratio C : O by mass and the likely formula (C = 12, O = 16) are: 1

(a) 1 : 2, CO_2 **(b)** 3 : 4, CO **(c)** 3 : 8, CO_2 **(d)** 3 : 16, CO_3

16. Two ions X^{2+} and Y^{2-} each have 18 electrons. The atomic numbers of X and Y are respectively: 1

(a) 16 and 20 **(b)** 20 and 16 **(c)** 19 and 17 **(d)** 18 and 18

17. **Assertion (A):** Mitochondria are called the powerhouses of the cell. 1
Reason (R): They release energy from food through cellular respiration and store it in the form of ATP.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Meristematic tissue is responsible for the growth of plants. 1
Reason (R): Its cells are actively dividing.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The species is the basic unit of classification. 1
Reason (R): Members of a species can interbreed to produce fertile offspring.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Offspring produced by asexual reproduction are genetically identical to the parent. 1
Reason (R): Asexual reproduction involves a single parent and no fusion of gametes.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. The average albedo of the Earth is 0.3. Calculate the radiation absorbed per square metre if the average incident is 342 W/m squared. 2

22. Why does a log-log or straight-line graph help in checking a mathematical relationship? 2

23. Explain why the thermometer reading stays constant during distillation of pure water. 2

24. Write the electronic configurations of Na^{+} and F^{-} and say what they have in common. 2

25. State the reason why the first shell is filled with only two electrons. 2

26. A body is dropped from rest and also another thrown down with 10 m/s from the same height. Compare the time of fall if h = 20 m (g = 10). 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. An atom has 5 more neutrons than protons and its mass number is 31. Find Z, n and the electronic configuration. 3

28. A ball rolls down an incline from rest and covers 2 m in the 1st second. Find the acceleration and the distance in 4 s. 3

29. A bullet of mass 10 g moving at 400 m/s enters a wooden block and stops in 0.04 m. Find the average retarding force and the time of penetration. 3

30. A spring stores 4 J of energy when stretched by 2 cm. If the energy is proportional to the square of the extension, how much energy does it store when stretched by 4 cm? 3

31. A siren emits a sound of frequency 700 Hz. Find the time period, the wavelength in air (340 m/s), and the number of waves reaching a man in 5 s. 3

OR

A researcher wants to isolate pure mitochondria from liver cells. Describe the method in outline (centrifugation). 3

32. State the reasons for the huge numbers of mitochondria in the cardiac muscle cells. 3

OR

Compare the life cycles of a moss and a fern briefly, noting the dominant stage in each. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** A metro train in Delhi starts from rest at a station and accelerates uniformly at 0.5 m/s^2 for 40 s. It then moves at constant velocity for the next 60 s. 5

(i) Find the velocity of the train after 40 s.
(ii) Find the distance covered in the first 40 s.
(iii) Find the distance covered in the next 60 s.
(iv) Find the total distance covered in the 100 s and the average speed over this time.

34. **Case study.** A porter at a railway station pushes a loaded trolley of mass 20 kg on a level platform with a horizontal force of 50 N. The platform exerts a friction force of 10 N opposing the motion. Take $g = 10$ m/s^2. 5

(i) Find the weight of the trolley.
(ii) Find the net force on it.
(iii) Find its acceleration.
(iv) If the trolley starts from rest, find its velocity and the distance travelled after 5 s.

35. Explain why echoes cannot be heard in a small room, but multiple echoes are heard in a big hall, and design the dimensions needed for a hall so that reverberation time is reduced. 5

36. Explain cell specialisation and the differentiation of stem cells with examples from plants and animals, and discuss medical uses and ethical issues. 5

OR

Explain the importance of structural adaptations of tissues for survival of plants in different habitats, with examples from xerophytes, hydrophytes and mesophytes. 5

37. Explain the adaptations of plants to life on land from algae to angiosperms, listing the structures that appeared: roots, vascular tissue, seeds, flowers, fruits and their advantages. 5

OR

Discuss the causes and prevention of infertility and the techniques of assisted reproduction including IVF, with the social aspects. 5

38. Explain how deforestation affects the water cycle, the carbon cycle, the albedo and the biodiversity in a region with examples from the Amazon or the Western Ghats. 5

OR

Discuss the strengths and limitations of the scientific method with five points and two examples of discoveries involving chance and persistence. 5

END OF QUESTION PAPER **Total: 80 Marks**

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